

INF1004 Mathematics II

Practice Tutorial: Linear Systems & Matrices

SIT

Question 1: System of Linear Equations

Determine the solution to the system of linear equations below:

- $x + y + z = 1$
- $x - y - 2z = 1$
- $-3x - 3y - 3z = -3$

Question 2: Vector Equation of a Line

A Cartesian equation that describes a line is given by:

$$z = -y = 2x + 1$$

Determine the unit vector, $\hat{v}$, in the vector equation below that describes the same line:

$$\vec{r} = \vec{u} + t\hat{v}$$

Question 3: Eigendecomposition and SVD

Given the matrix A :

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

- 1 Determine the eigendecomposition of A .
- 2 Given that there is a square matrix B such that $B^T B = A$, determine the singular value decomposition (SVD) of matrix B .

Question 4: Upper Diagonal System

Compute the solution vector (x, y, z) for this equation system, which is already upper diagonal:

- $8x - 11y + 27z = 31$
- $6y - 13z = 18$
- $5z = 34$

Question 5: Orthogonal Unit Vector

Compute the vector (x, y, z) which:

- is orthogonal to both of the vectors $(3, -7, 10)$ and $(-2, 1, -4)$.
- has its very first component x being positive.
- has unit magnitude.

Question 6: Eigenvectors

Find the eigenvector $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ for the eigenvalue $c = 5$ for the matrix:

$$A = \begin{pmatrix} -0.5 & -2 & 2.5 \\ 1.5 & -1 & -1.5 \\ -5.5 & -2 & 7.5 \end{pmatrix}$$

Question 7: Eigenvalues

Find the eigenvalues of matrix S :

$$S = \begin{pmatrix} -4 & 2 \\ 3 & 6 \end{pmatrix}$$